Department of Information Technology

Academic Year 2025 - 2026 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT
Course Code & Title: CS3452 & Theory of Computation
Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit V / **Recursive and Recursively enumerable Languages, Properties**
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO16
- **Activity Chosen:** One Minute Paper
- **Justification:**

  In unit V, the concept Recursive and recursively enumerable languages, Properties is an important topic. The One Minute Paper is a quick formative assessment technique used at the end of a class to capture students' understanding in a concise manner. The activity provides instant insight into what students have understood and what remains unclear. It helps the instructor quickly gauge the effectiveness of the teaching and identify learning gaps.
- **Time Allotted for the Activity:** 5 Minutes

- **Details of the Implementation:**
  - On the day the instructor explained the topic Recursive and Recursively enumerable languages, properties.
  - The time allotted for the topic is 5 Minutes. The students during the activity as shown in Figure 1
  - The students were asked to recall the concepts of Recursive and recursively enumerable languages and its properties
  - Next, the students were asked to write down their doubts on a sheet of paper.
  - The papers were collected and corrected as shown in Figure 2.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO5	2	3	1	2	2	1	2	2	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO5	2	3	1	2	2	1	2	2	2	2	2
Justification for correlation	Apply basic Knowledge and fundamentals of Recursive and recursively enumerable languages, properties.	The students will be able to identify the need for Recursive and recursively enumerable languages in the context of complex engineering problems	Able to design solutions for Recursive and recursively enumerable languages Finite Automata and Regular Expressions.	Able to analyze and apply Recursive and recursively enumerable languages concept	Able to follow ethics for solving the problem	Able to implement the problem individually	Able to communicate the concept Recursive and recursively enumerable languages.	Ability to recognize the need for Recursive and recursively enumerable languages in real world applications	Exhibit knowledge in Recursive and recursively enumerable languages for application development	Exhibit knowledge in Recursive and recursively enumerable languages of IoT, Cloud Computing and Data Analytics to provide IT solutions	Exhibit knowledge in Recursive and recursively enumerable languages in software projects to deliver a quality product.

- **Images / Screenshot of the practice:**

Figure 1: One Minute Paper Activity

Theory of Computation

One – Minute Paper

Recursive and Recursively Enumerable Language and its properties

From this topic. I understood that a recursive language is a language for which a turing machine always halts and gives a definite answer accept or reject for every input. This means membership can be decided completely whereas recursively enumerable language accepts valid inputs but may run forever for invalid ones. Every recursive language is recursively enumerable but not all recursively enumerable languages are recursive

Recursive languages always halt for every input are closed under operations like union, intersection and complement.

Recursively enumerable languages may not halt for some inputs and every recursive language belongs to this class but not vice versa.

Theory of Computation

Activity - One Minute Paper

Recursive and Recursively Enumerable Language and its properties

In this topic. I understood the difference between recursive and recursively enumerable language.

A recursive language is one for which a Turning machine always halts and gives a correct answer for every input.

On the other hand a recursively enumerable language is one where the Turning machine halts and accepts string that belongs to the language. but not halt for strings that do not belong.

I also learned that all recursive languages are recursively enumerable but not all recursively enumerable language are ~~a~~ recursive. This concept helps in understanding the limits of computation and decidability in theory of computation.

Figure 2: Student Responses Ms. Karthiga G and S. Kavya

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - Students expressed that they were able to recall the concepts of Recursive and Recursively Enumerable Languages, along with their key properties, more effectively through the One Minute Paper activity.
    - They shared that the activity helped reinforce their understanding of decidability, recognizability, and closure properties discussed in the previous class.
    - Many students reported that writing brief responses encouraged them to think independently and organize their understanding of the concepts clearly.
  - ❖ ***Benefit of the practice:***
    - Encouraged active reflection on core concepts such as recursive languages, recursively enumerable languages, and their properties.
    - Helped in identifying specific areas where students faced confusion, particularly in distinguishing between recursive and recursively enumerable languages.
    - Promoted self-assessment and enhanced engagement in higher-order thinking skills through concise written responses.
  - ❖ ***Challenges faced in implementation***
    - Some students found it difficult to clearly differentiate between recursive and recursively enumerable languages and their respective properties.
    - Additional explanations, examples, and individual guidance were provided to support slow learners, which improved their conceptual clarity.

References:

- https://ohioline.osu.edu/
- https://www.nureva.com/blog/education/15-active-learning-activities-to-energize-your-next-college-class
- https://www.ritrjpm.ac.in/images/B.Tech-IT/2025-2026/4.One%20minute%20paper.pdf
- https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/VA_CS3452_OneMinutePaper.pdf

Signature of Faculty Member **HOD**